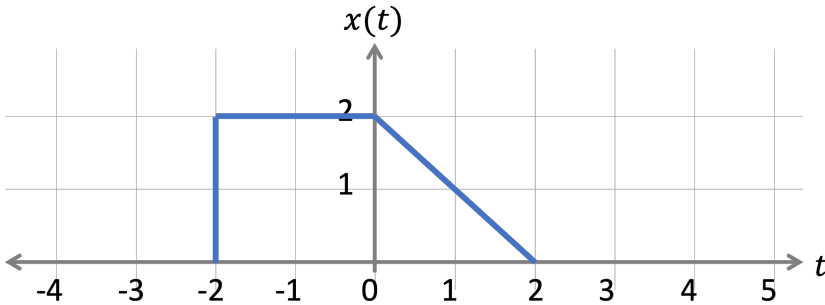


ECE 3337: Signals & Systems Analysis Class Worksheet 1

1. (70 points) Consider the following waveform $x(t)$:



Plot the following waveforms as a function of t :

- $y_1(t) = x(0.25t)$. Does this transformation compress or expand the waveform in time?
- $y_2(t) = x(2t)$. Does this transformation compress or expand the waveform in time?
- $y_3(t) = x(-2t)$
- $y_4(t) = x(2t + 3)$ using Approach #1 (time scale followed by time shift)
- $y_5(t) = x(2t + 3)$ using Approach #2 (time shift followed by time scale). Does your answer match the one with Approach #1? It should!
- $y_6(t) = x(-2t + 3)$ using Approach #1
- $y_7(t) = x(2t - 3)$ using Approach #2

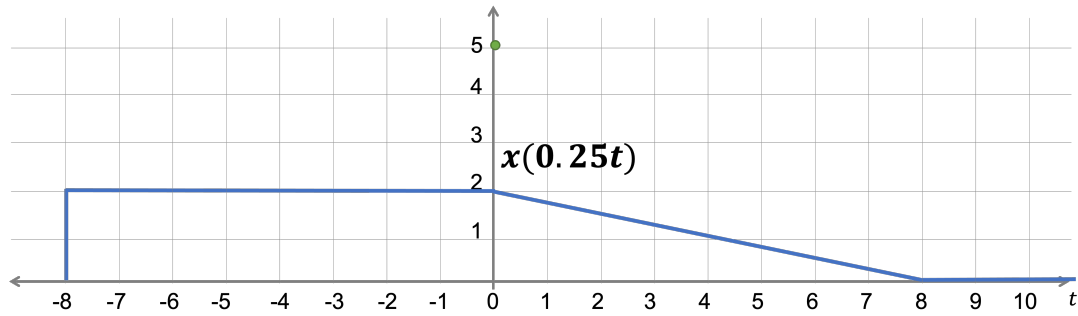
2. (30 points) Plot the following signals on the same chart:

- $x_1(t) = \sin(t)$
- $x_2(t) = \sin(2t)$
- $x_3(t) = \sin(4t - \frac{\pi}{2})$

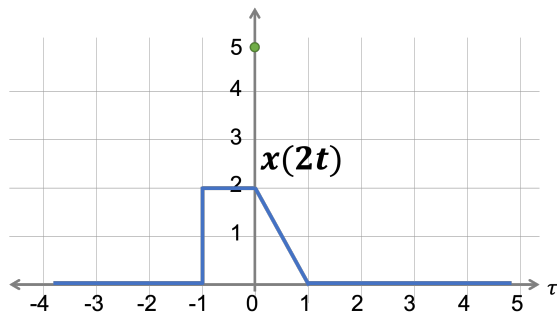
10 points for each part in Q1 and Q2. 3 points for trying and 7 for full correct answer.

Solutions

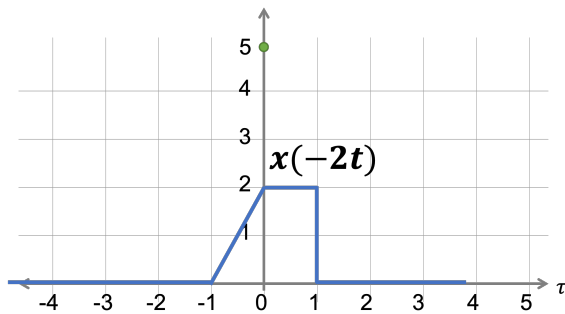
1 (a)



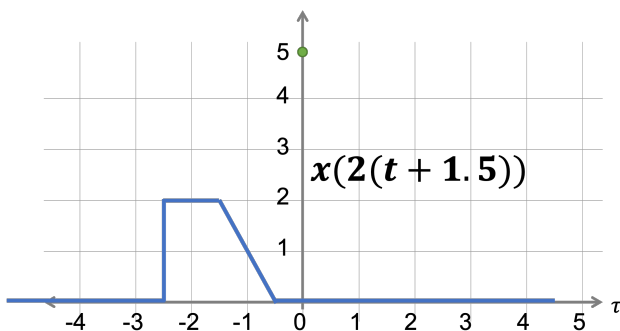
(b)



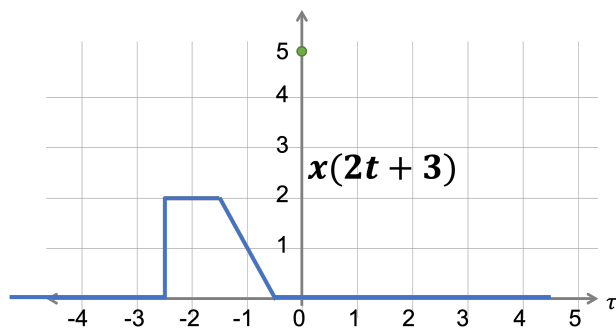
(c)



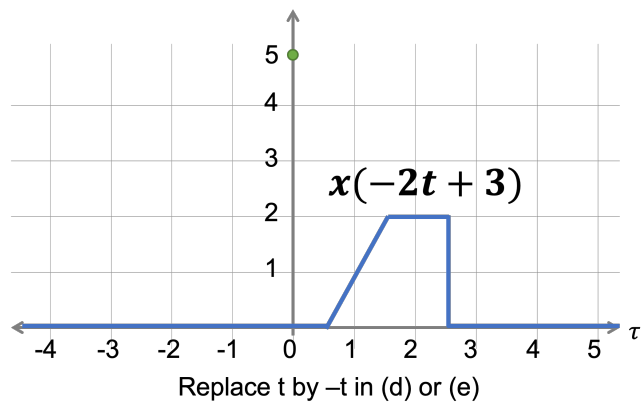
(d)



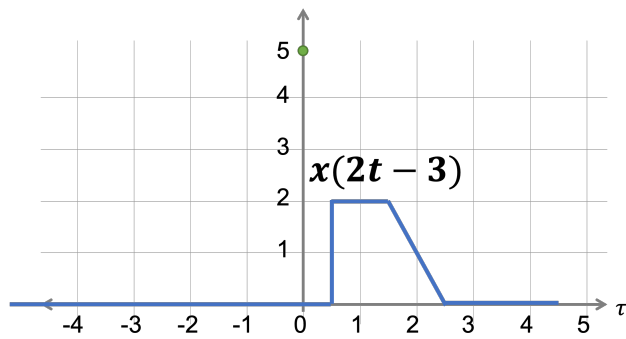
(e)



(f)



(g)



②

